

Algebraic Geometry II

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In Algebraic Geometry I, we studied the classical approach of varieties in affine or projective space. While this is very geometric, there remain issues: Firstly, one needs the base field K to be algebraically closed: otherwise, Hilbert's Nullstellensatz fails, and the correspondence between geometric and algebraic objects breaks down (e.g. $X^2 + 1 = 0$ has no solutions in $\mathbb{R}$). In fact, we would like to not only look at non-algebraically closed fields, but more generally at algebraic sets over arbitrary rings, to study, for instance, algebraic equations over the integers using geometric methods.

Another issue of classical varieties comes from intersections: for example, take $K = \mathbb{C}$ and look at $V(y - x^2) \cap V(y) \subseteq \mathbb{A}^2$. One can compute this intersection to be $V(x^2, y) = V(x, y)$, which corresponds to the single point $(0, 0)$. However, we would like to distinguish the algebraic sets $V(x^2, y)$ and $V(x, y)$, since the former carries the extra information of "multiplicity". Algebraically, this corresponds to non-reduced K -algebras. If we had such a notion of multiplicity, we could prove more powerful theorems. For instance, Bézout's theorem simply reads $|X \cap Y| = \deg X \deg Y$, where the points in $X \cap Y$ are counted with multiplicity.

To resolve these problems, and at the same time introduce a much more general and powerful framework, Grothendieck introduced the language of schemes. In the classical setting, note that the points of an affine variety $X \subseteq \mathbb{A}^n$ correspond to the maximal ideals of $A(X)$. In the new setting, we will look at all prime ideals of $A(X)$.

For an arbitrary ring A , we can give $\text{Spec } A$ a Zariski topology, and define regular functions on open subsets of $\text{Spec } A$. This is all similar to the classical ideas, but involves more technical objects. This will define the notion of an *affine scheme*.

1 Sheaves

Definition 1.1. Let X be a topological space. A *presheaf* $\mathcal{F}$ of abelian groups on X consists of an abelian group $\mathcal{F}(U)$ for each open subset $U \subseteq X$ and a group homomorphism $\rho_{U,V} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ for any $V \subseteq U$ open, such that $\rho_{U,U} = \text{id}_{\mathcal{F}(U)}$ and $\rho_{V,W} \circ \rho_{U,V} = \rho_{U,W}$ for any $W \subseteq V \subseteq U$ open.

In other words, a presheaf is a contravariant functor from the category of open subsets of X to the category of abelian groups. The homomorphisms $\rho_{U,V}$ are called *restriction maps*. An element $s \in \mathcal{F}(U)$ is called a *section* of $\mathcal{F}$ on U . If $V \subseteq U$ is open, we will usually write $s|_V := \rho_{U,V}(s)$.

Remark 1.2. Analogously one defines presheaves of sets, groups, rings, R -modules etc.

Example 1.3. Let X be a topological space, and let $\mathcal{F}(U) := \{f : U \rightarrow \mathbb{R} \mid f \text{ continuous}\}$, with $\rho_{U,V}$ the usual restriction of functions. Similarly, if X carries additional structure, one can look at conformal, differentiable or holomorphic functions on X .

Let X be a quasi-projective variety over a field K . Then the regular functions on X form a presheaf $\mathcal{O}_X(U) = \{f : U \rightarrow K \mid f \text{ regular}\}$.

In all examples, the presheaf $\mathcal{F}$ satisfies the following two properties:

Definition 1.4. A presheaf $\mathcal{F}$ on X is a *sheaf* if the following two conditions hold for any open $U \subseteq X$ and every open cover $U = \bigcup_{i \in I} U_i$:

- (i) If $s, t \in \mathcal{F}(U)$ with $s|_{U_i} = t|_{U_i}$ for all $i \in I$, then $s = t$.
- (ii) If $s_i \in \mathcal{F}(U_i)$ for all $i \in I$ with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$, then there exists $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$ for all $i \in I$.

In fact, the section s in (ii) is unique by (i).

Remark 1.5. The examples in 1.3 are all sheaves. For an example of a presheaf that is not necessarily a sheaf, let $1 \neq A$ be any abelian group, X a topological space, and $\mathcal{F}(U) := A$ for all opens $U \subseteq X$. If X is not connected, this is not a sheaf. To rectify this, give A the discrete topology, and define the *constant sheaf* $\underline{A}$ by

$$\underline{A}(U) := \text{Hom}(U, A) = \{f : U \rightarrow A \mid f \text{ locally constant}\}$$

Definition 1.6. Let $\mathcal{F}$ be a presheaf on X , and let $x \in X$. Then the *stalk* of $\mathcal{F}$ at x is defined as the direct limit

$$\mathcal{F}_x := \varinjlim_{x \in U} \mathcal{F}(U) = \{(U, s) \mid x \in U, s \in \mathcal{F}(U)\} / \sim,$$

where $(U, s) \sim (V, t)$ if there exists an open $x \in W \subseteq U \cap V$ with $s|_W = t|_W$.

Remark 1.7. $\mathcal{F}_x$ is an abelian group with operation $[(U, s)] + [(V, t)] = [(U \cap V, s|_{U \cap V} + t|_{U \cap V})]$.

Definition 1.8. Let X be a topological space. A morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ of (pre)sheaves on X is a family of morphisms $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ for all opens $U \subseteq X$, such that for all opens $V \subseteq U$, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi_U} & \mathcal{G}(U) \\ \downarrow \rho_{UV} & & \downarrow \rho_{UV} \\ \mathcal{F}(V) & \xrightarrow{\varphi_V} & \mathcal{G}(V) \end{array}$$

φ is an isomorphism if there exists an inverse morphism $\psi : \mathcal{G} \rightarrow \mathcal{F}$.

Remark 1.9. In other words, a morphism of (pre)sheaves is a natural transformation of the two functors.

Proposition 1.10. Let $x \in X$. A morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ of presheaves on X induces a morphism $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ of stalks, given explicitly by $[(U, f)] \mapsto [(U, \varphi(U)(f))]$.

Proof. This follows directly from the universal property of direct limits. $\square$

Proposition 1.11. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of schemes. Then φ is an isomorphism if and only if $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is an isomorphism for every $x \in X$.

Note that this fails for presheaves, and hence gives some motivation for the conditions in the definition of a sheaf.

Proof. If φ is an isomorphism, then so is φ_x (more precisely $(\varphi_x)^{-1} = (\varphi_x)^{-1}$) by functoriality of direct limits.

Conversely, assume that φ_x is an isomorphism for every $x \in X$, and let $U \subseteq X$ be open. Assume $s \in \ker \varphi_U$. For any $x \in U$, we have $\varphi_x([(U, s)]) = [U, \varphi(U)(s)] = 0$, so by injectivity of φ_x we get $[(U, s)] = 0$ in $\mathcal{F}_x$. That is, $s = 0$ on some neighbourhood $x \in U_x \subseteq U$. Then $U = \bigcup_{x \in U} U_x$, so by 1.4(1) we get $s = 0 \in \mathcal{F}(U)$. Hence φ_U is injective.

For surjectivity, let $t \in \mathcal{G}(U)$. For any $x \in U$, φ_x is surjective, so let $[(U_x, s_x)] \in \mathcal{F}_x$ be a preimage of $[(U, t)] \in \mathcal{G}_x$, wlog $U_x \subseteq U$. Then $U = \bigcup_{x \in X} U_x$ and $s_x \in \mathcal{F}(U_x)$. To check the gluing condition, let $x, y \in U$ and compute

$$\varphi_{U_x \cap U_y}(s_x|_{U_x \cap U_y}) \stackrel{1.8}{=} (\varphi_{U_x}(s_x))|_{U_x \cap U_y} = t|_{U_x}|_{U_x \cap U_y} \stackrel{1.1}{=} t|_{U_x \cap U_y} = \varphi_{U_x \cap U_y}(s_y|_{U_x \cap U_y}),$$

thus by injectivity $s_x|_{U_x \cap U_y} = s_y|_{U_x \cap U_y}$. Hence by 1.4(2) we obtain an element $s \in \mathcal{F}(U)$ s.t. $s|_{U_x} = s_x$ for every $x \in X$. It remains to check that $\varphi_U(s) = t$. But this holds by 1.4(1) applied to $\mathcal{G}$, since by construction $\varphi_U(s)|_{U_x} = \varphi_{U_x}(s|_{U_x}) = \varphi_{U_x}(s_x) = t|_{U_x}$. Therefore, φ_U is surjective, hence an isomorphism. $\square$

Definition 1.12 (Naive¹). Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves. We say that φ is injective/surjective if for every $x \in X$, $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is injective/surjective. In particular, if $\mathcal{F}, \mathcal{G}$ are sheaves, then by proposition 1.11, φ is an isomorphism if and only if it is injective and surjective.

Definition 1.13. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves on X . We define the following presheaves:

$$\begin{aligned}\ker(\varphi)(U) &:= \ker(\varphi_U), \\ \operatorname{im}(\varphi)^{\operatorname{pre}}(U) &:= \operatorname{im}(\varphi_U) \\ \operatorname{coker}(\varphi)^{\operatorname{pre}}(U) &:= \operatorname{coker}(\varphi_U)\end{aligned}$$

Remark 1.14. In each case, one easily checks that these assignments define presheaves. If $\mathcal{F}, \mathcal{G}$ are sheaves, then $\ker \varphi$ is a sheaf as well (Exercise), but in general $\operatorname{im} \varphi$ and $\operatorname{coker} \varphi$ are not sheaves.

Proposition 1.15 (Sheafification). *Let $\mathcal{F}$ be a presheaf on X . Then there is a unique (up to unique isomorphism) sheaf $\mathcal{F}^+$, called sheafification of $\mathcal{F}$, and a morphism $\theta : \mathcal{F} \rightarrow \mathcal{F}^+$ with the following universal property: For any sheaf $\mathcal{G}$ and morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ of presheaves, there exists a unique morphism $\tilde{\varphi} : \mathcal{F}^+ \rightarrow \mathcal{G}$ making the following diagram commute:*

$$\begin{array}{ccc}\mathcal{F} & \xrightarrow{\varphi} & \mathcal{G} \\ \searrow \theta & & \nearrow \tilde{\varphi} \\ & \mathcal{F}^+ & \end{array}$$

Moreover, θ induces isomorphisms $\theta_x : \mathcal{F}_x \rightarrow \mathcal{F}_x^+$ for every $x \in X$. If $\mathcal{F}$ is already a sheaf, then θ is an isomorphism.

Proof. We construct $\mathcal{F}^+$ as follows:

$$\mathcal{F}^+(U) := \left\{ s : U \rightarrow \prod_{x \in U} \mathcal{F}_x \mid s(x) \in \mathcal{F}_x \text{ for all } x \in U, \text{ and } (*) \right\},$$

where $(*)$ is the following extra condition: For every $x \in U$ there exists an open neighbourhood $x \in V \subseteq U$ and $t \in \mathcal{F}(V)$ with $s(y) = t_y := [(V, t)]$ for all $y \in V$. Further, set $\theta_U(t)(x) = t_x$ for all $U \subseteq X$ open, $t \in \mathcal{F}(U)$ and $x \in U$. One checks that this $\mathcal{F}^+, \theta$ have the required properties:

It is clear that $\mathcal{F}^+$ is a sheaf, since it is constructed via functions with local properties, and that θ is a morphism of presheaves, since $(t|_W)_x = t_x$ for $x \in W \subseteq V$, $t \in \mathcal{F}(V)$. Let $x \in X$ and $[(V, t)] \in \ker \theta_x$. Up to shrinking V , we have $\theta_V(t) = 0$. In particular, $[(V, t)] = \theta_V(t)(x) = 0$. Let $[(V, s)] \in \mathcal{F}_x^+$. By $(*)$ wlog there is $t \in \mathcal{F}(V)$ s.t. $s(y) = t_y$ for all $y \in V$. Hence $\theta_V(t) = [(V, s)]$. If $\mathcal{F}$ is a sheaf, then θ is thus an isomorphism by proposition 1.11.

Finally, we check the universal property: Let $\mathcal{G}$ be a sheaf, and $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves. By the previous paragraph, it suffices to give a morphism $\tilde{\varphi} : \mathcal{F}^+ \rightarrow \mathcal{G}^+ \cong \mathcal{G}$. Let $U \subseteq X$ be open and let $\tilde{\varphi}(s)(y) = \varphi_y(s(y))$. It is easy to check that this is a well-defined morphism of sheaves, makes the diagram commute, and is the only morphism with this property. The details are left as an exercise. $\square$

Example 1.16. Let X be a topological space such that for every $x \in X$ there is a neighbourhood $x \in U_x \subsetneq X$, and let $0 \neq A$ an abelian group.

¹will be corrected in 1.21

- (i) Let $\mathcal{F}(X) = A$ and $\mathcal{F}(U) = 0$ otherwise. This is clearly not a sheaf. By the assumption on X we have $\mathcal{F}_x = 0$ for all $x \in X$, so $\mathcal{F}^+ = 0$.
- (ii) Let $\mathcal{F}$ be the constant presheaf with value A as in remark 1.5. From the universal property one sees that $\mathcal{F}^+ = \underline{A}$ is the constant sheaf.

Definition 1.17. A sub-(pre)sheaf of a (pre)sheaf $\mathcal{F}$ on X is a (pre)sheaf $\mathcal{F}'$ s.t. for every open $U \subseteq X$ one has $\mathcal{F}'(U) \subseteq \mathcal{F}(U)$, and for every $V \subseteq U$ open, $\rho'_{UV} = \rho_{UV}|_{\mathcal{F}'(U)}$. In other words, the inclusions $\mathcal{F}'(U) \hookrightarrow \mathcal{F}(U)$ have to form a morphism of (pre)sheaves.

If $\mathcal{F}' \subseteq \mathcal{F}$ is a sub-presheaf, then for every $x \in X$ we have $\mathcal{F}'_x \subseteq \mathcal{F}_x$.

Proposition 1.18. Let $\mathcal{F}$ be a sheaf and $\mathcal{F}'$ be a subsheaf. Then $\mathcal{F}' = \mathcal{F}$ if and only if $\mathcal{F}'_x = \mathcal{F}_x$ for all $x \in X$.

Proof. Follows directly from proposition 1.11. □

Example 1.19. If $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of presheaves, then $\ker \varphi$ and $\operatorname{im} \varphi^{\text{pre}}$ are sub-presheaves of $\mathcal{F}$ resp. $\mathcal{G}$.

Exercise 1.20. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves. Show that $(\ker \varphi)_x = \ker(\varphi_x)$ and $(\operatorname{im} \varphi^{\text{pre}})_x = \operatorname{im}(\varphi_x)$.

Definition 1.21. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves. We say that φ is injective if and only if $\ker \varphi = 0$. For presheaves, we say φ is surjective if $\operatorname{coker} \varphi^{\text{pre}} = 0^2$. If $\mathcal{F}, \mathcal{G}$ are sheaves, define the image and cokernel sheaves via $\operatorname{im} \varphi := (\operatorname{im} \varphi^{\text{pre}})^+$ and $\operatorname{coker} \varphi := (\operatorname{coker} \varphi^{\text{pre}})^+$. Then $\operatorname{im} \varphi$ can be identified with a subsheaf of $\mathcal{G}$ (see next proposition). We say that φ is surjective as a morphism of sheaves if $\operatorname{im} \varphi = \mathcal{G}$, equivalently $\operatorname{coker} \varphi = 0$.

Proposition 1.22. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves. Then $\operatorname{im} \varphi$ can be seen as a subsheaf of $\mathcal{G}$. Moreover, $\operatorname{im} \varphi = \mathcal{G} \iff \varphi_x$ is surjective for all $x \in X$.

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Proof. We have $\operatorname{im} \varphi \subseteq \mathcal{G}^+ \cong \mathcal{G}$. The second part follows directly from proposition 1.11. □

Remark 1.23. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of (pre)sheaves. Then φ is injective if and only if φ_U is injective for all $U \subseteq X$ open, that is $\ker \varphi = 0$. If φ is injective, then also $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is injective, but the converse is only true for sheaves.

φ is a surjective morphism of presheaves if φ_U is surjective for all $U \subseteq X$ open, that is $\operatorname{im} \varphi^{\text{pre}} = \mathcal{G}$. Again, if φ is surjective, then so are all φ_x , but the converse is not true in general (even for sheaves).

If $\mathcal{F}, \mathcal{G}$ are sheaves, φ is a surjective morphism of sheaves if and only if $\operatorname{im} \varphi = \mathcal{G}$, if and only if φ_x is surjective for all $x \in X$.

Example 1.24. (i) Let $\mathcal{F}, \mathcal{G}$ be (pre)sheaves on X . Then $(\mathcal{F} \oplus \mathcal{G})(U) := \mathcal{F}(U) \oplus \mathcal{G}(U)$ is again a (pre)sheaf.

(ii) Let A be an abelian group, X a topological space and $x \in X$. The sheaf

$${}^x\mathcal{F}(U) := \begin{cases} A, & \text{if } x \in U, \\ 0 & \text{otherwise} \end{cases}$$

is called the *skyscraper sheaf*. One calculates ${}^x\mathcal{F}_y = A$ if $y \in \overline{\{x\}}$, and ${}^x\mathcal{F}_y = 0$ otherwise. In particular, if X is T_1 (that is, points are closed), then ${}^x\mathcal{F}_y \neq 0$ only for $x = y$.

²Note that this is not equivalent to φ_x being surjective for all $x \in X$.

- (iii) Let X be a connected T_1 -space, $x \neq z \in X$. Let $0 \neq A$ be an abelian group, and consider the natural map $\varphi : \underline{A} \rightarrow {}^x\mathcal{F} \oplus {}^z\mathcal{F}$. One sees that φ is surjective as a morphism of sheaves, but not as a morphism of presheaves.

Definition 1.25. Let $f : X \rightarrow Y$ be a continuous map of topological spaces.

- (i) Let $\mathcal{F}$ be a sheaf on X . The *direct image* of $\mathcal{F}$ is the sheaf $f_*\mathcal{F}$ on Y defined by

$$f_*\mathcal{F}(U) := \mathcal{F}(f^{-1}(U)).$$

- (ii) Let $\mathcal{G}$ be a sheaf on Y . The *inverse image* $f^{-1}\mathcal{G}$ of $\mathcal{G}$ is defined as the sheafification of the presheaf

$$U \mapsto \varinjlim_{V \supseteq f(U)} \mathcal{G}(V).$$

2 Affine Schemes

In Algebraic Geometry I, we showed an equivalence of categories between affine algebraic sets over K , and finitely generated reduced K -algebras, for K algebraically closed. We want to generalize this to allow any ring A on the algebraic side. A natural generalization of the classical setting is to consider $\text{Spec } A$, and to endow this set with topological and algebraic information.

Let A be a ring (always commutative with 1).

Definition 2.1. $\text{Spec } A := \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ prime ideal}\}$. If $I \subseteq A$ is any ideal, let $V(I) := \{\mathfrak{p} \in \text{Spec } A \mid I \subseteq \mathfrak{p}\}$. (Note that there is a canonical bijection $V(I) \cong \text{Spec } A/I$.) For $S \subseteq A$ any subset, we write $V(S) := V(\langle S \rangle)$.

Lemma 2.2. Let $I, J \subseteq A$ be ideals.

- (i) $\sqrt{I} := \{f \in A \mid f^n \in I \text{ for some } n \geq 1\} = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p}$.
- (ii) $I \subseteq \sqrt{I}$ and $V(I) = V(\sqrt{I})$.
- (iii) If $I \subseteq J$, then $V(J) \subseteq V(I)$.
- (iv) If $V(J) \subseteq V(I)$ if and only if $\sqrt{I} \subseteq \sqrt{J}$.

Proof. (i) was proven in Commutative Algebra. (ii), (iii), and one direction of (iv) are clear. So assume $V(J) \subseteq V(I)$, then $\sqrt{I} = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p} \subseteq \bigcap_{\mathfrak{p} \in V(J)} \mathfrak{p} = \sqrt{J}$ by (i). $\square$

Lemma 2.3. Let $I, J, \{I_\alpha\}_{\alpha \in I} \subseteq A$ be ideals.

- (i) $V(IJ) = V(I \cap J) = V(I) \cup V(J)$.
- (ii) $V(\sum_\alpha I_\alpha) = \bigcap_\alpha V(I_\alpha)$.

Proof. (i) For a prime ideal $\mathfrak{p} \subseteq A$, we have $IJ \subseteq \mathfrak{p}$ if and only if $I \subseteq \mathfrak{p}$ or $J \subseteq \mathfrak{p}$, if and only if $I \cap J \subseteq \mathfrak{p}$.

(ii) Exercise. $\square$

Definition 2.4. By lemma 2.3 (and the observations $V(A) = \emptyset$, $V(0) = \text{Spec } A$), the collection $\{V(I) \mid I \subseteq A \text{ ideal}\}$ forms the set of closed sets of a topology on $\text{Spec } A$, the so-called *Zariski topology*.

Remark 2.5. Not every point of $\text{Spec } A$ is closed. More precisely, let $x \in \text{Spec } A$. Then $\overline{\{x\}} = V(x)$, so $\{x\}$ is closed if and only if x is a maximal ideal.

Example 2.6. (i) If A is an integral domain, then $0 \in \text{Spec } A$ satisfies $\overline{\{0\}} = \text{Spec } A$. In particular, every open in $\text{Spec } A$ contains $\{0\}$. Such a point is called a *generic point* of $\text{Spec } A$.

(ii) Let $\mathbb{A}^1 := \text{Spec } K[x]$. Then $\mathbb{A}^1$ contains the generic point (0) , plus closed points corresponding to normalised irreducible polynomials in $K[x]$. If K is algebraically closed, every irreducible polynomial is linear, so $\mathbb{A}^1$ looks very similar to affine space in the classical setting, the only difference being the additional generic point.

(iii) $\text{Spec } \mathbb{Z} = \{(0)\} \cup \{(p) \mid p \text{ prime}\}$

(iv) Let $\mathbb{A}^2 := \text{Spec } K[x, y]$ with K algebraically closed. It contains the generic point (0) , closed points $(x - a, y - b)$ as in the classical settings, but also (principal) prime ideals corresponding to curves. Let (F) be such an ideal, then

$$\overline{\{(F)\}} = \{(F)\} \cup \{(x - a, y - b) \mid F(a, b) = 0\}.$$

Lemma 2.7. (i) If A is an integral domain, then $\text{Spec } A$ is irreducible. (The converse fails e.g. for $K[x]/(x^2)$). If A is reduced and $\text{Spec } A$ is irreducible, then A is an integral domain.

- (ii) $\text{Spec } A$ is not connected if and only if there exists $0, 1 \neq e \in A$ with $e^2 = e$, if and only if $A \cong A_1 \times A_2$ with $A_1, A_2 \neq 0$.

Proof. (i) Assume that $\text{Spec } A$ is irreducible and A is reduced but not an integral domain. Let $a * b = 0$ with $0 \neq a, b \in A$. Then $\text{Spec } A = V(0) = V(a) \cup V(b)$. If $V(a) = \text{Spec } A$, then $a \in \sqrt{0} = 0$ by assumption, contradiction. Hence $\text{Spec } A$ is reducible. The rest is left as an exercise. $\square$

Definition 2.8. Let $f \in A$. Then $D(f) := \text{Spec } A \setminus V(f) = \{\mathfrak{p} \in \text{Spec } A \mid f \notin \mathfrak{p}\}$ is called a *distinguished* or *standard open* subset of $\text{Spec } A$.

Remark 2.9. One has $D(f) \cap D(g) = D(fg)$. The set $\{D(f) \mid f \in A\}$ forms a basis of the Zariski topology on $\text{Spec } A$: Let $U = \text{Spec } A \setminus V(I)$ be open and $\mathfrak{p} \in U$. Since $\mathfrak{p} \notin V(I)$, $I \not\subseteq \mathfrak{p}$. Hence choose $f \in I \setminus \mathfrak{p}$, which satisfies $\mathfrak{p} \in D(f) \subseteq U$.

We want to construct a sheaf of "regular functions" on $\text{Spec } A$. Recall from the classical setting, that if $X \subseteq \mathbb{A}^n$ was an affine algebraic set and $U \subseteq X$ open, we defined $\mathcal{O}_X(U)$ as the set of all functions $f : U \rightarrow K$ that locally are rational functions, and we proved $\mathcal{O}_X(D(f)) \cong A(X)_f$ and $\mathcal{O}_{X,x} \cong \varinjlim_{D(f) \ni x} A(X)_f \cong A(X)_{\mathfrak{m}_x}$. This motivates us to define a sheaf $\mathcal{O}_X$ of rings on $\text{Spec } A$ s.t. $\mathcal{O}_X(D(f)) \cong A_f$. Note that if $D(g) \subseteq D(f)$, then $g^m = fa$ for some $a \in A, m \geq 1$ by lemma 2.2, so there is a natural map $A_f \rightarrow A_g$. Since we know what the stalks are supposed to look like, we can proceed as in the proof of sheafification 1.15.

Definition 2.10. Let $X = \text{Spec } A$. The structure sheaf $\mathcal{O}_X$ is defined as

$$\mathcal{O}_X(U) := \left\{ s : U \rightarrow \prod_{\mathfrak{p} \in U} A_{\mathfrak{p}} \mid s(\mathfrak{p}) \in A_{\mathfrak{p}} \text{ and } (*) \right\},$$

where $(*)$ is the extra condition that for every $\mathfrak{p} \in U$ there exists an open neighbourhood $V \subseteq U$ and $a, f \in A$ such that $s(\mathfrak{q}) = \frac{a}{f}$ in $A_{\mathfrak{q}}$ for every $\mathfrak{q} \in V$. By construction it is clear that $\mathcal{O}_X$ is a sheaf of rings on $\text{Spec } A$.

Proposition 2.11. For $\mathfrak{p} \in \text{Spec } A$ we have $\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}$, and for $f \in A$, $\mathcal{O}_X(D(f)) \cong A_f$. In particular, $\mathcal{O}_X(X) \cong A$.

Proof. Let $\mathfrak{p} \in \text{Spec } A$ and set $\varphi : \mathcal{O}_{X,\mathfrak{p}} \rightarrow A_{\mathfrak{p}}$, $[(U, f)] \mapsto f(\mathfrak{p})$. This is a well-defined homomorphism of rings. Let $\frac{a}{s} \in A_{\mathfrak{p}}$ with $a, s \in A, s \notin \mathfrak{p}$. That is, $\mathfrak{p} \in D(s)$, so $[(D(s), \frac{a}{s})]$ is a well-defined preimage of $\frac{a}{s}$. Conversely, let $[(U, f)] \in \ker \varphi$. By shrinking U , we may assume $f(\mathfrak{q}) = \frac{a}{s} \in A_{\mathfrak{q}}, a, s \in A$, for all $\mathfrak{q} \in U$. Then $0 = \varphi([(U, f)]) = \frac{a}{s} \in A_{\mathfrak{p}}$, hence $s'a = 0$ for some $s' \notin \mathfrak{p}$, i.e. $\mathfrak{p} \in D(s')$. By shrinking U further, if necessary, we may assume $U \subseteq D(s')$. Then $f = \frac{a}{s} = \frac{s'a}{s's} = 0$. Therefore, φ is an isomorphism $\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}$.

Now let $f \in A$. Define $\psi : A_f \rightarrow \mathcal{O}_X(D(f)), \alpha \mapsto (t_{\alpha} : \mathfrak{q} \mapsto \alpha)$. This is clearly a well-defined homomorphism of rings. Let $\alpha = \frac{a}{f^n} \in \ker \psi$. That is, $\frac{a}{f^n} = 0$ in $A_{\mathfrak{q}}$ for every $\mathfrak{q} \in D(f)$. This means that for every $\mathfrak{q} \in D(f)$ there exists $s_{\mathfrak{q}} \notin \mathfrak{q}$ with $s_{\mathfrak{q}}a = 0$. Consider the ideal $I = \{b \in A \mid ba = 0\}$ (called the annihilator of a). Then for every $\mathfrak{q} \in D(f)$, $s_{\mathfrak{q}} \in I \setminus \mathfrak{q}$, i.e. $I \not\subseteq \mathfrak{q}$. Hence $V(I) \subseteq \text{Spec } A \setminus D(f) = V(f)$. By lemma 2.2 it follows that $f^m \in I$ for some $m > 0$. Then $\alpha = \frac{a}{f^n} = \frac{af^m}{f^{n+m}} = 0$.

For surjectivity, let $t \in \mathcal{O}_X(D(f))$. By definition, there is an open cover $D(f) = \bigcup_i V_i$ s.t. $t|_{V_i} = \frac{a_i}{s_i}$ for some $a_i, s_i \in A$ with $V_i \subseteq D(s_i)$. By remark 2.9 wlog $V_i = D(h_i)$. Now $V(h_i) \supseteq V(s_i)$, so again by lemma 2.2, $h_i \in \sqrt{(s_i)}$, i.e. $h_i^{n_i} = s_i c_i$ for some $n_i > 0, c_i \in A$. It follows that $t|_{D(h_i)} = \frac{a_i}{s_i} = \frac{a_i c_i}{h_i^{n_i}}$. Replacing $h_i^{n_i}$ by h_i (since $D(h_i) = D(h_i^{n_i})$) and $a_i c_i$ by a_i , we simply write $t|_{D(h_i)} = \frac{a_i}{h_i}$.

Observe that there exists a finite subcover $D(f) = \bigcup_{i=1}^k D(h_i)$ of $\bigcup_i D(h_i)$. Indeed, $D(f) \subseteq \bigcup_i D(h_i)$ implies $V(f) \supseteq \bigcap_i V(h_i) = V(\sum_i (h_i))$ by lemma 2.3. By lemma 2.2, $f^n \in \sum_i (h_i)$ for some $n > 0$. But this means f^n is a *finite* linear combination of the h_i 's. Taking only those h_i which appear in this sum and retracing the argument backwards yields the claim.

For $i \neq j$, we have $\frac{a_i h_j}{h_i h_j} = t|_{D(h_i) \cap D(h_i h_j)} = t|_{D(h_j) \cap D(h_i h_j)} = \frac{a_j h_i}{h_j h_i}$ as functions on $D(h_i h_j)$. By the injectivity of ψ (for $D(h_i h_j)$) proved above, even $\frac{a_i h_j}{h_i h_j} = \frac{a_j h_i}{h_j h_i}$ as elements of $A_{h_i h_j}$, i.e. $(h_i h_j)^n (a_i h_j - a_j h_i) = 0$ in A , for some n . Since there are only finitely many sets in the open cover, we can choose n big enough such that this equality holds for all i, j .

Replacing h_i by h_i^{n+1} and a_i by $h_i^n a_i$, we still have $t|_{D(h_i)} = \frac{a_i}{h_i}$, and now $h_j a_i = h_i a_j$ for all i, j . As before, $f \in \sqrt{(h_1, \dots, h_k)}$, so we can write $f^m = \sum_i b_i h_i$ for some $m > 0$ and $b_i \in A$. Let $a := \sum_i b_i a_i$. We claim that $\psi(\frac{a}{f^m}) = t$. Indeed, for every j we have $f^m a_j = \sum_i b_i h_i a_j = \sum_i b_i a_i h_j = a h_j$, i.e. $\frac{a}{f^m} = \frac{a_j}{h_j} \in A_q$ for every $q \in D(h_j)$. $\square$

Lecture 7
May 4, 2026

Having now associated to a ring A the pair $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$, we next want to define morphisms between these objects. We let us motivate by the following

Remark 2.12. Recall from classical algebraic geometry that if X, Y are quasi-projective algebraic sets, a map $\varphi : X \rightarrow Y$ was defined to be a morphism if φ is continuous, and for every open $U \subseteq Y$ and $f \in \mathcal{O}_Y(U)$, its preimage $\varphi^* f = f \circ \varphi$ is regular, i.e. lies in $\mathcal{O}_X(\varphi^{-1}(U))$. This yielded compatible homomorphisms of K -algebras $\mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(\varphi^{-1}(U))$, in other words a morphism of sheaves $\mathcal{O}_Y \rightarrow \varphi_* \mathcal{O}_X$.

- Definition 2.13.** (i) A *ringed space* is a pair $(X, \mathcal{O}_X)$, where X is a topological space and $\mathcal{O}_X$ a sheaf of rings on X .
- (ii) A morphism of ringed spaces $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ is a pair $(f, f^\#)$, where $f : X \rightarrow Y$ is a continuous map, and $f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$ is a morphism of sheaves.
- (iii) A *locally ringed space* is a ringed space $(X, \mathcal{O}_X)$ such that all stalks $\mathcal{O}_{X,x}$, $x \in X$ are local rings.
- (iv) A morphism of locally ringed spaces $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces such that for every $x \in X$, the map $f_x^\# : \mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$ is a local homomorphism of rings (where a morphism of rings $\varphi : (A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$ of local rings is called *local* if $\varphi^{-1}(\mathfrak{n}) = \mathfrak{m}$). The map $f_x^\#$ is defined as the compositum

$$\mathcal{O}_{Y,f(x)} \xrightarrow{f_{f(x)}^\#} (f_* \mathcal{O}_X)_{f(x)} = \varinjlim_{f^{-1}(U) \ni x} \mathcal{O}_X(f^{-1}(U)) \xrightarrow{\text{incl}_*} \varinjlim_{U \ni x} \mathcal{O}_X(U) = \mathcal{O}_{X,x},$$

where the second map is induced by inclusion of directed sets.

Remark 2.14. Let A be a ring. The pair $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$ is a locally ringed space by proposition 2.11.

- Definition 2.15.** (i) Let $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ and $(g, g^\#) : (Y, \mathcal{O}_Y) \rightarrow (Z, \mathcal{O}_Z)$ be morphisms of (locally) ringed spaces. Then their composition is defined as $(g, g^\#) \circ (f, f^\#) := (g \circ f, (g \circ f)^\#)$, with $(g \circ f)^\# := g_* f^\# \circ g^\# : \mathcal{O}_Z \rightarrow (g \circ f)_* \mathcal{O}_X$.
- (ii) An isomorphism of (locally) ringed spaces is a morphism with a two-sided inverse morphism.

A morphism $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is an isomorphism if and only if $f : X \rightarrow Y$ is a homeomorphism and $f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$ is an isomorphism of sheaves. (Exercise)

Lemma 2.16. Let $\varphi : A \rightarrow B$ be a morphism of rings. Then $f : \text{Spec } B \rightarrow \text{Spec } A$, $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$ is continuous.

Proof. f is well-defined since the preimage of a prime ideal is again prime. Let $V(I) \subseteq \operatorname{Spec} A$. Then $f^{-1}(V(I)) = \{\mathfrak{p} \in \operatorname{Spec} B \mid \varphi^{-1}(\mathfrak{p}) \in V(I)\} = V(\varphi(I))$ is closed. $\square$

Remark 2.17. Let $\varphi : A \rightarrow B$ be a ring morphism and $X = \operatorname{Spec} B, Y = \operatorname{Spec} A$. Then φ induces a morphism of locally ringed spaces $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$, where f is the map of lemma 2.16, and $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ is defined on $U \subseteq Y$ as

$$f_U^\# : \mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(f^{-1}(U)), \quad t \mapsto (\mathfrak{p} \mapsto \varphi_{\mathfrak{p}}(t(f(\mathfrak{p}))),$$

where $\varphi_{\mathfrak{p}} : A_{f(\mathfrak{p})} \rightarrow B_{\mathfrak{p}}$ is the natural morphism induced by the universal property of localisation. Note that if $t|_V = \frac{a}{s}$, then $f^\#(t)|_{f^{-1}(V)} = \frac{\varphi(a)}{\varphi(s)}$, so $f^\#$ is indeed a well-defined morphism of sheaves.

We check that $(f, f^\#)$ is a morphism of locally ringed spaces, i.e. that $f_{\mathfrak{p}}^\# : \mathcal{O}_{Y, f(\mathfrak{p})} \rightarrow (f_*\mathcal{O}_X)_{f(\mathfrak{p})} \rightarrow \mathcal{O}_{X, \mathfrak{p}}$ is a local morphism of rings. By definition, this map is given by $[(U, t)] \mapsto [(f^{-1}(U), f^\#(t))]$, and hence by construction the diagram

$$\begin{array}{ccc} \mathcal{O}_{Y, f(\mathfrak{p})} & \xrightarrow{f_{\mathfrak{p}}^\#} & \mathcal{O}_{X, \mathfrak{p}} \\ \downarrow \cong & & \downarrow \cong \\ A_{f(\mathfrak{p})} & \xrightarrow{\varphi_{\mathfrak{p}}} & B_{\mathfrak{p}} \end{array}$$

commutes, where the vertical isomorphisms come from proposition 2.11. Since $\varphi_{\mathfrak{p}}$ is local, hence so is $f_{\mathfrak{p}}^\#$.

As was the case in classical algebraic geometry, we have

Proposition 2.18. *Let A, B be rings, and $X = \operatorname{Spec} B, Y = \operatorname{Spec} A$. Then the construction above yields a bijection*

$$\operatorname{Hom}_{\mathbf{Ring}}(A, B) \xrightarrow{\cong} \operatorname{Hom}_{\mathbf{LRS}}((X, \mathcal{O}_X), (Y, \mathcal{O}_Y)).$$

(In other words, the functor Spec is fully faithful.)

Proof. Let $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of locally ringed spaces. Then $f_Y^\# : \mathcal{O}_Y(Y) \cong A \rightarrow B \cong f_*\mathcal{O}_X(Y)$, we claim that this construction is inverse to the one constructed in remark 2.17. Given $\varphi : A \rightarrow B$, let $(f, f^\#)$ be the induced morphism of locally ringed spaces. Then $f_Y^\# = \varphi$ (under the canonical identifications) by construction.

Conversely, let $(f, f^\#)$ be a morphism of locally ringed spaces, and let $(g, g^\#)$ be the morphism constructed from $\varphi := f_Y^\#$ as above. Let $\mathfrak{p} \in X$. We have a commutative diagram

$$\begin{array}{ccc} B \cong \mathcal{O}_X(X) & \xleftarrow{f_Y^\#} & \mathcal{O}_Y(Y) \cong A \\ \downarrow & & \downarrow \\ B_{\mathfrak{p}} \cong \mathcal{O}_{X, \mathfrak{p}} & \xleftarrow{f_{\mathfrak{p}}^\#} & \mathcal{O}_{Y, f(\mathfrak{p})} \cong A_{f(\mathfrak{p})} \end{array}$$

Since $f_{\mathfrak{p}}^\#$ is a local morphism, the preimage of $\mathfrak{p}B_{\mathfrak{p}}$ is the image of $f(\mathfrak{p})$ in $A_{f(\mathfrak{p})}$. By commutativity of the diagram, this equals $\varphi^{-1}(\mathfrak{p})$, i.e. $g = \varphi^{-1} = f$.

By the universal property of localizations, there is a unique map $A_{f(\mathfrak{p})} \rightarrow B_{\mathfrak{p}}$ making the above diagram commute. But both $f_{\mathfrak{p}}^\#$ and $\varphi_{\mathfrak{p}}$ do, so they are equal. By the construction in remark 2.17, this immediately implies $f^\# = g^\#$. $\square$

Definition 2.19. An *affine scheme* is a locally ringed space isomorphic to $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$ for some ring A .

Remark 2.20. Immediately from the definition and proposition 2.18 it follows that the functor $\text{Spec} : \text{Ring} \rightarrow \text{AffSch}$, $A \mapsto \text{Spec } A$, $\varphi \mapsto (f, f^\#)$ as in remark 2.17 is an (anti-)equivalence of categories.

Definition 2.21. A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ such that for every $x \in X$ there exists a neighbourhood $x \in U \subseteq X$ such that $(U, \mathcal{O}_X|_U)$ is an affine scheme.

By abuse of notation, we will often write X instead of $(X, \mathcal{O}_X)$ and f instead of $(f, f^\#)$. Given a scheme X , we will write $\text{Sp}(X)$ for the underlying topological space.

Lecture 9
May 11, 2026

Lemma 2.22. Let A be a ring and $X = \text{Spec } A$. Let $g \in A$ and consider $D(g) \subseteq \text{Spec } A$. Then $(D(g), \mathcal{O}_X|_{D(g)}) \cong \text{Spec } A_g$ is again an affine scheme.

Proof. Exercise. □

Proposition 2.23. Let X be a scheme and $U \subseteq X$ be open. Then $(U, \mathcal{O}_X|_U)$ is again a scheme.

Proof. $(U, \mathcal{O}_X|_U)$ is clearly a locally ringed space. We only need to check that locally it is an affine scheme, so wlog $X = \text{Spec } A$ is affine. Let $U = \bigcup_i D(g_i)$ be an open cover by standard opens, then each $(D(g_i), \mathcal{O}_X|_{D(g_i)}) = (D(g_i), \mathcal{O}_X|_U|_{D(g_i)})$ is an affine scheme by the previous lemma, so the result follows immediately. □

Definition 2.24. (i) Let X be a scheme and $U \subseteq X$ be open. Then $(U, \mathcal{O}_X|_U)$ is called an *open subscheme* of X . If moreover U is an affine scheme, then we say that U is an *affine open subset/subscheme* of X . Observe that by lemma 2.22, affine open subsets of X form a basis of the topology of X .

(ii) Consider the natural morphism $i_U : U \rightarrow X$ of schemes, where $(i_U^\#)_V : \mathcal{O}_X(V) \rightarrow i_{U*}\mathcal{O}_X|_U(V) = \mathcal{O}_X(U \cap V)$ is the restriction. An *open immersion* is a morphism of schemes $Y \rightarrow X$ which induces an isomorphism of Y with some open subscheme $U \subseteq X$, that is, it factors through i_U .

(iii) Let $f : X \rightarrow Y$ be a morphism of schemes and $U \subseteq X$ be open. Then we consider the restriction of f on U , $f|_U := f \circ i_U : U \rightarrow Y$.

Proposition 2.25 (Glueing of morphisms). Let X, Y be schemes and $X = \bigcup_{i \in I} U_i$ be an open cover. Then the map

$$\text{Hom}_{\text{Sch}}(X, Y) \rightarrow \prod_{i \in I} \text{Hom}_{\text{Sch}}(U_i, Y), \quad f \mapsto (f|_{U_i})_i$$

is injective, and has image consisting of those $(f_i)_i$ with $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all $i, j \in I$.

Proof. Let $(f, f^\#) : X \rightarrow Y$ be given. It is clear that f can be recovered uniquely from the $(f|_{U_i})_i$. Let $V \subseteq Y$ be open. Since $f_V^\#(s) \in \mathcal{O}_X(f^{-1}(V))$ is uniquely determined by its restrictions to $(f^{-1}(V) \cap U_i)_i$ by the sheaf condition, we can recover $f^\#$ as well.

It is clear that any family $(f_i)_i$ in the image satisfies the given condition. Conversely, suppose $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ holds for all $i, j \in I$. On the level of topological spaces the claim is clear: For $x \in X$ set $f(x) := f_i(x)$ for any i with $x \in U_i$. Now let $U \subseteq Y$ be open, $s \in \mathcal{O}_Y(U)$. The family of sections $t_i := (f_i)_U^\#(s) \in \mathcal{O}_{U_i}(U \cap U_i) = \mathcal{O}_X(U \cap U_i)$ glue to a unique section $f_U^\#(s)$, since $t_i|_{U_i \cap U_j} = (f_i)_{U_i \cap U_j}^\#(s) = (f_j)_{U_i \cap U_j}^\#(s) = t_j|_{U_i \cap U_j}$ by assumption. By uniqueness, this construction respects addition and multiplication, so $f_U^\#$ is a ring morphism. It is clear that $f^\#$ is a morphism of sheaves, and that $f|_{U_i} = f_i$, hence $f_x^\#$ is local for all $x \in X$, and $(f, f^\#)$ is a morphism of schemes as required. □

3 Properties and some Constructions of Schemes

Definition 3.1. A scheme X is called *irreducible/connected* if its underlying topological space is.

Definition 3.2. A scheme X is called *reduced/integral* if for every open $U \subseteq X$, the ring $\mathcal{O}_X(U)$ is reduced/an integral domain.

Remark 3.3. An affine scheme $X = \operatorname{Spec} A$ is irreducible if and only if the nilradical of A is prime. In particular, if A is an integral domain then X is irreducible. Furthermore, X is reduced/integral if and only if A is.